

Account Demo Program

Java Lab Record

1. Aim

To develop a Java program to demonstrate basic class and object concepts by creating an Account class with methods for deposit, withdrawal, interest calculation, and displaying account details.

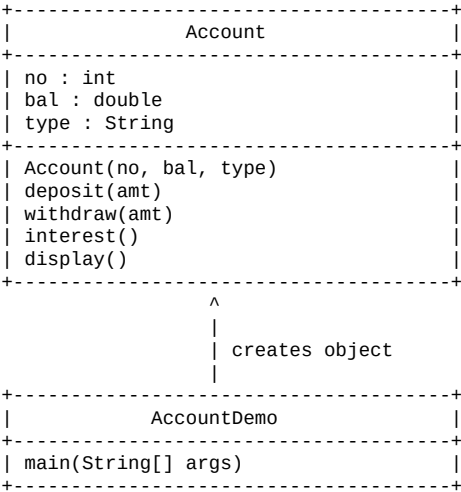
2. Step-by-Step Procedure

1. Create a class named Account.
2. Declare account number, balance, and account type as data members.
3. Create a parameterized constructor to initialize the account details.
4. Create the deposit() method to add an amount to the account balance.
5. Create the withdraw() method to withdraw an amount when sufficient balance is available.
6. Display an insufficient balance message when the withdrawal amount is greater than the balance.
7. Create the interest() method to calculate interest at 4% of the current balance.
8. Create the display() method to print account number, account type, and balance.
9. Create the AccountDemo class containing the main() method.
10. Create an Account object with account number 1001, balance 10000, and type Savings.
11. Display the initial account details.
12. Deposit 2000 and withdraw 3000.
13. Calculate and display the interest.
14. Display the final account details.
15. Stop the program.

3. Algorithm

1. Start.
2. Create an Account object with account number, balance, and account type.
3. Display the initial account details.
4. Deposit 2000 into the account.
5. Check whether 3000 can be withdrawn from the available balance.
6. If sufficient balance is available, withdraw 3000; otherwise display Insufficient Balance.
7. Calculate interest as $\text{balance} \times 0.04$.
8. Display the calculated interest.
9. Display the updated account details.
10. Stop.

4. UML Class Diagram



5. Code

```
class Account {
    int no;
    double bal;
    String type;

    Account(int no,double bal,String type) {
        this.no=no; this.bal=bal; this.type=type;
    }

    void deposit(double amt) {
        bal+=amt;
        System.out.println("Deposited: "+amt);
    }

    void withdraw(double amt) {
        if(amt<=bal) {
            bal-=amt;
            System.out.println("Withdrawn: "+amt);
        } else
            System.out.println("Insufficient Balance");
    }

    void interest() {
        System.out.println("Interest: "+bal*0.04);
    }

    void display() {
        System.out.println("No: "+no);
        System.out.println("Type: "+type);
        System.out.println("Balance: "+bal);
    }
}

public class AccountDemo {
    public static void main(String[] args) {
        Account a=new Account(1001,10000,"Savings");

        a.display();
        a.deposit(2000);
        a.withdraw(3000);
        a.interest();
        a.display();
    }
}
```

6. Sample Output

The following output is obtained by executing the given AccountDemo program.

```
--- Account Demo ---
```

```
No: 1001
```

```
Type: Savings
```

```
Balance: 10000.0
```

```
Deposited: 2000.0
```

```
Withdrawn: 3000.0
```

```
Interest: 360.0
```

```
No: 1001
```

```
Type: Savings
```

```
Balance: 9000.0
```